Distribuciones de Probabilidad

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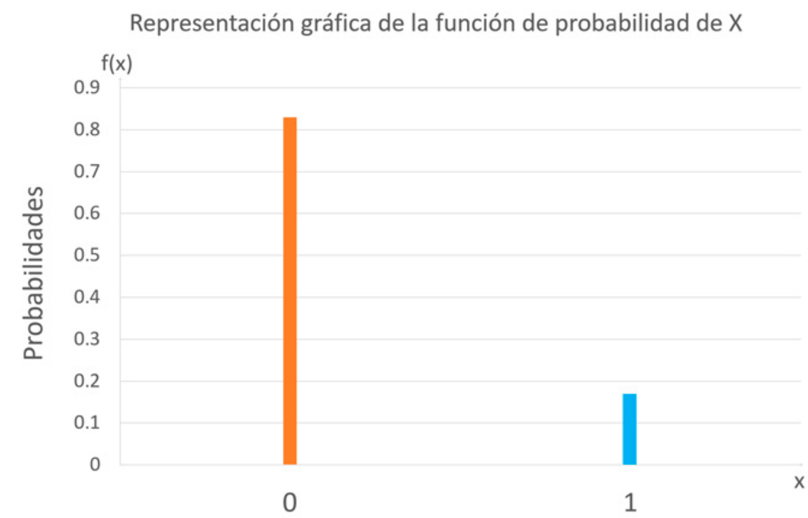
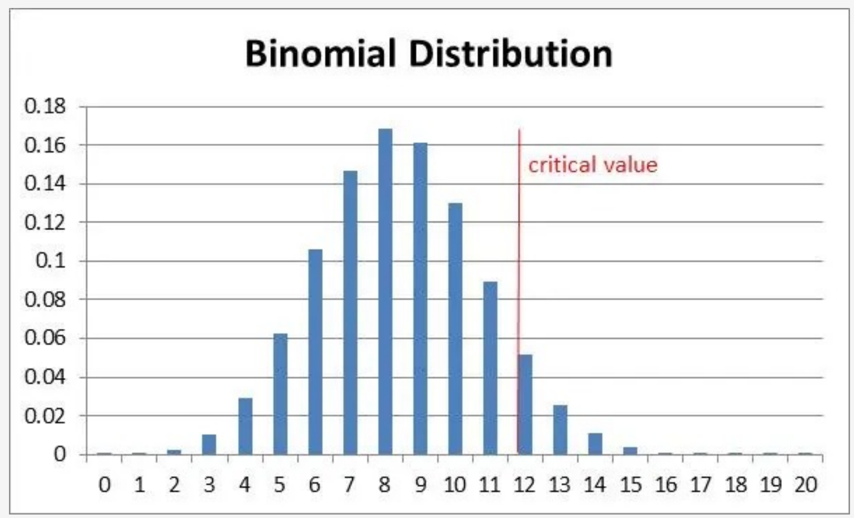
Distribuciones Discretas

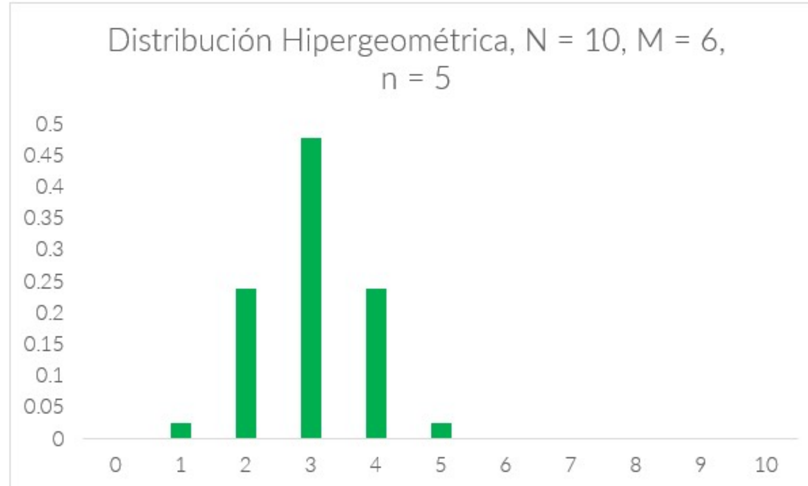
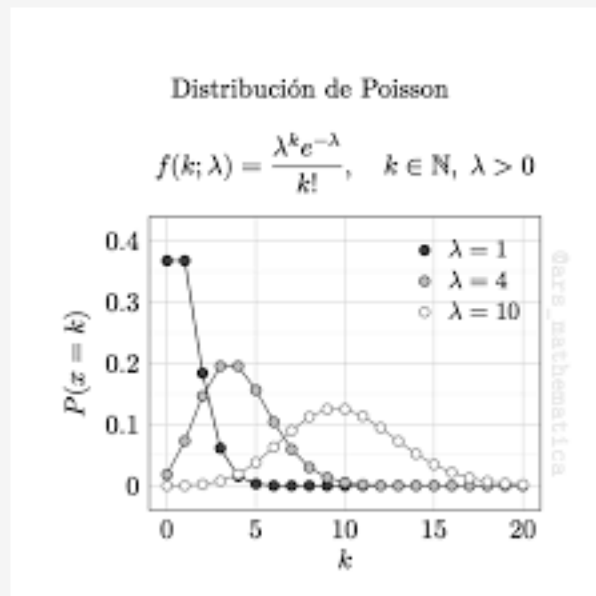
Distribución	**Notación**	**Función de Probabilidad**	**Función Acumulada**	**Momentos**	**FGM**
Bernoulli	$X \sim Ber(p)$	$P(X=x) = p^x(1-p)^{1-x}$	$F(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$	$E(X) = p$ $Var(X) = p(1-p)$	$M_X(t) = 1 - p + pe^t$
Binomial	$X \sim Bin(n,p)$	$\binom{n}{x} p^x (1-p)^{n-x}$	$\sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k}$	$E(X) = np$ $Var(X) = np(1-p)$	$(1-p+pe^t)^n$
Hipergeométrica	$X \sim H(N,K,n)$	$P(X=x) = \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$F(x) = \sum_{k=0}^{x} \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$	$E(X) = n\frac{K}{N}$ $Var(X) = n\frac{K}{N}\left(1-\frac{K}{N}\right)\frac{N-n}{N-1}$	$\sum_x e^{tx} P(X=x)$
Poisson	$X \sim Pois(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$\sum_{k=0}^{x} \frac{e^{-\lambda}\lambda^k}{k!}$	$E(X) = \lambda$ $Var(X) = \lambda$	$e^{\lambda(e^t-1)}$
Geométrica	$X \sim Geom(p)$	$(1-p)^{x-1}p, \quad x = 1, 2, \ldots$	$1-(1-p)^x$	$E(X) = \frac{1}{p}$ $Var(X) = \frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$

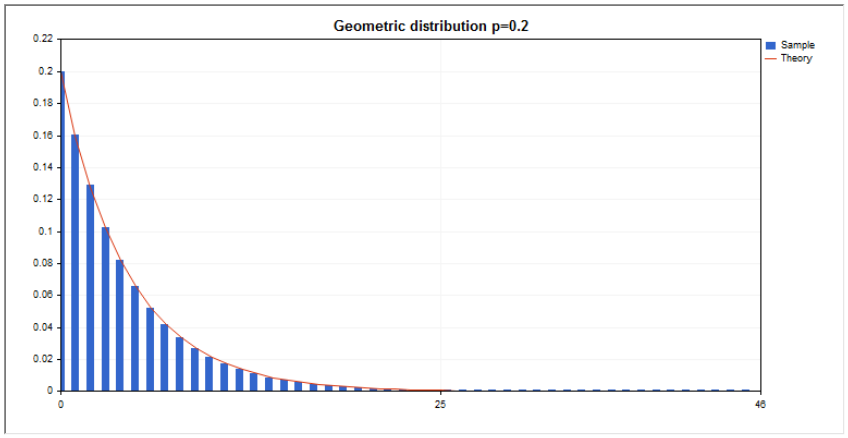
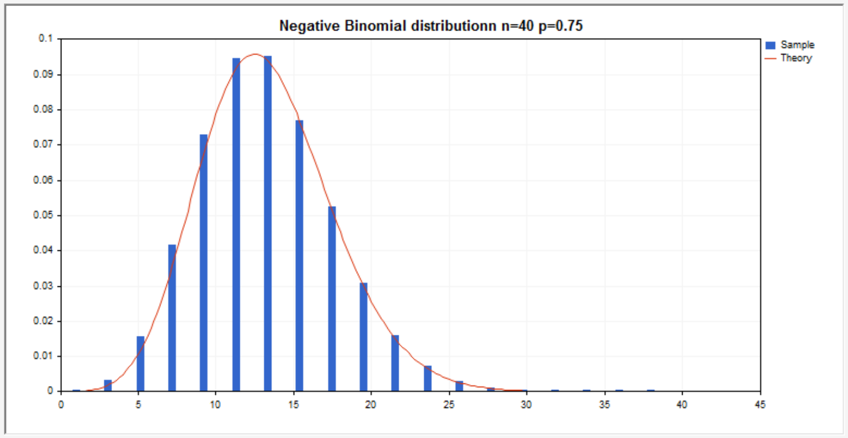
Binomial Negativa	$X \sim NB(r,p)$	$P(X=x)=\binom{x+r-1}{x}(1-p)^x p^r$	$F(x)=\sum_{k=0}^{x}\binom{k+r-1}{k}(1-p)^k p^r$	$E(X)=\frac{r(1-p)}{p}$ $Var(X)=\frac{r(1-p)}{p^2}$	$\left(\frac{p}{1-(1-p)e^t}\right)^r$

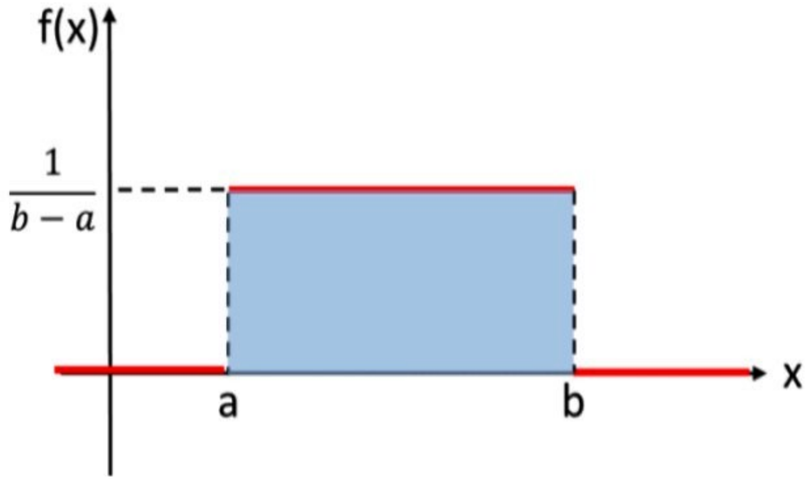
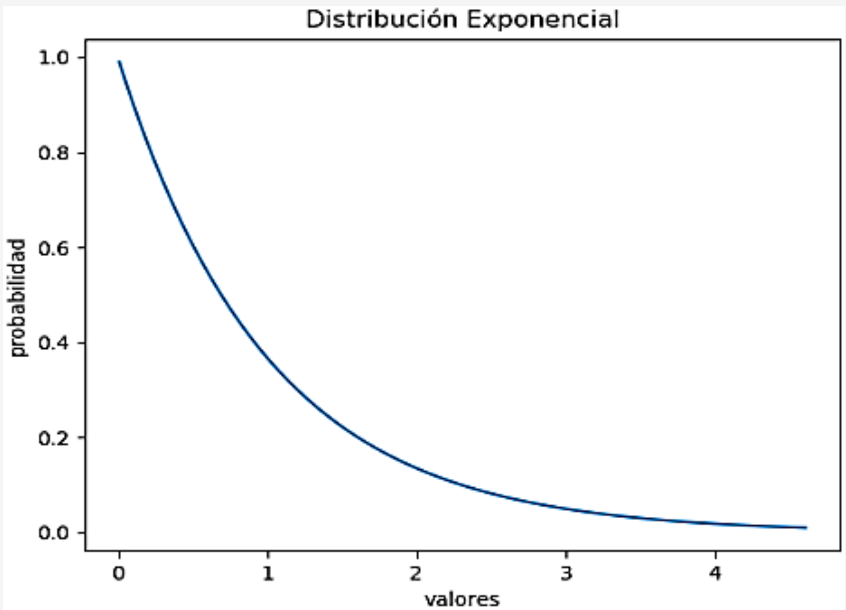
Distribuciones Continuas

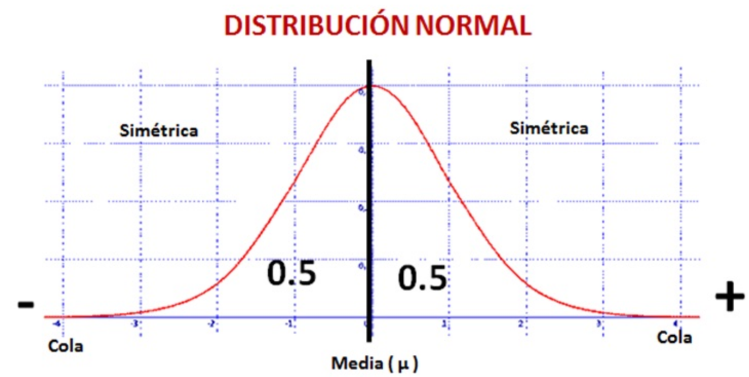
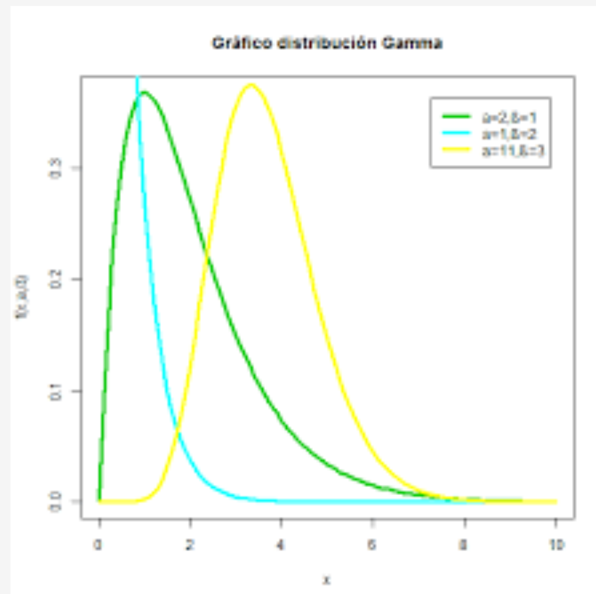
Distribución	**Notación**	**Función de Densidad**	**Función Acumulada**	**Momentos**	**FGM**
Uniforme	$X \sim U(a,b)$	$\frac{1}{b-a}, \quad a \leq x \leq b$	$\frac{x-a}{b-a}, \quad a \leq x \leq b$	$E(X)=\frac{a+b}{2}$ $Var(X)=\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}, \quad t \neq 0$
Exponencial	$X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}, \quad x > 0$	$1-e^{-\lambda x}, \quad x > 0$	$E(X)=\frac{1}{\lambda}$ $Var(X)=\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}, \quad t < \lambda$
Normal	$X \sim N(\mu,\sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$	$\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{x} e^{-\frac{(t-\mu)^2}{2\sigma^2}}\,dt$	$E(X)=\mu$ $Var(X)=\sigma^2$	$e^{\mu t+\frac{\sigma^2 t^2}{2}}$
Gamma	$X \sim \Gamma(\alpha,\lambda)$	$\frac{\lambda^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\lambda x}, \quad x > 0$	$\frac{\gamma(\alpha,\lambda x)}{\Gamma(\alpha)}, \quad x > 0$	$E(X)=\frac{\alpha}{\lambda}$ $Var(X)=\frac{\alpha}{\lambda^2}$	$\left(\frac{\lambda}{\lambda-t}\right)^\alpha, \quad t < \lambda$
Beta	$X \sim Beta(\alpha,\beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1$	$I_x(\alpha,\beta)$	$E(X)=\frac{\alpha}{\alpha+\beta}$ $Var(X)=\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\int_0^1 e^{tx}f(x)\,dx$
Chi-cuadrada	$X \sim \chi^2(\nu)$	$\frac{1}{2^{\nu/2}\Gamma(\nu/2)}x^{\nu/2-1}e^{-x/2}, \quad x > 0$	$\frac{\gamma\left(\frac{\nu}{2},\frac{x}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)}, \quad x > 0$	$E(X)=\nu$ $Var(X)=2\nu$	$(1-2t)^{-\nu/2}, \quad t < \frac{1}{2}$

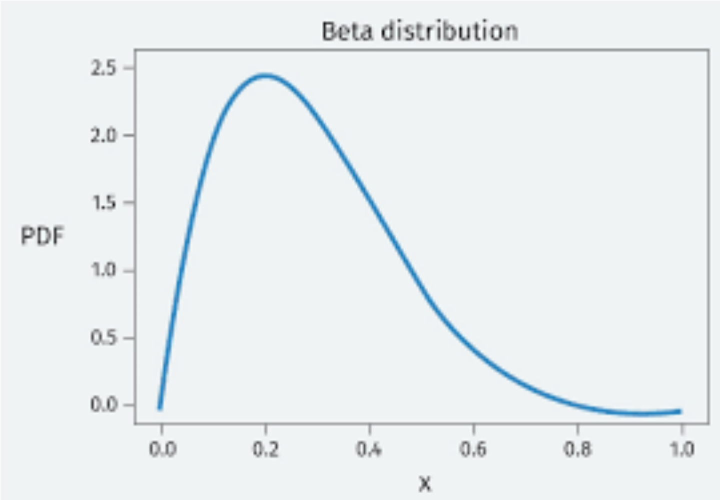
Distribución	Gráfica
Bernoulli	
Binomial	

Distribución	Gráfica
Hipergeométrica	
Poisson	

Distribución	Gráfica
Geométrica	
Binomial Negativa	

Distribución	Gráfica
Uniforme	
Exponencial	

Distribución	Gráfica
Normal	
Gamma	

Distribución	Gráfica
Beta	

Distribución	Gráfica
Chi-cuadrada